LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Safety and emergency devices for retraction and extension, typical u/c circuits
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Safety and emergency devices for retraction and extension, typical u/c circuits,
5. Specific Objectives	17.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of typical u/c circuit diagrams; Instructor supervision required for safety device identification
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary functions of the landing gear system?
(b) Name two safety devices used to prevent accidental gear retraction.
(c) How does the emergency extension system differ from normal retraction?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Typical alighting gear construction • Safety devices for retraction and extension • Emergency extension mechanisms • Typical u/c circuit components • Hydraulic and mechanical interlocks	L & Ds	PPT, IB	15 min
(b)	• Analysis of u/c circuit diagrams • Identification of safety valves • Operation of emergency extension • Role of uplocks and downlocks • System pressure monitoring	L & Ds	PPT, PDF	20 min
(c)	• Review of safety protocols • Troubleshooting common failures • Summary of circuit logic • Preparation for shock absorber study • Q&A session	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the function of the uplock mechanism in the landing gear circuit.
(b) Describe the sequence of events during an emergency gear extension.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between normal and emergency landing gear extension procedures to be discussed in LP 12.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Shock observers, types, brief descriptions, operation of oleos	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min